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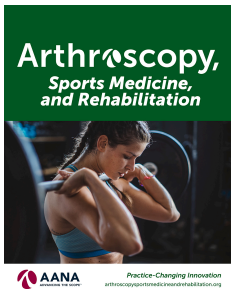

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PII: S2666-061X(24)00072-5

DOI: https://doi.org/10.1016/j.asmr.2024.100954

Reference: ASMR 100954

To appear in: *Arthroscopy, Sports Medicine, and Rehabilitation*

Received Date: 2 February 2024

Revised Date: 1 May 2024

Accepted Date: 9 May 2024

Please cite this article as: Quinn M, Painter DF, Ahn BJ, Byrne RA, Testa EJ, Albright A, Tabaddor RR, DeFroda SF, NFL Quarterbacks with Ulnar Collateral Ligament Injuries Have High Return to Play Rates, but Older Players Have Inferior Post-Injury Performance, *Arthroscopy, Sports Medicine, and Rehabilitation* (2024), doi: https://doi.org/10.1016/j.asmr.2024.100954.

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NFL Quarterbacks with Ulnar Collateral Ligament Injuries Have High Return to Play Rates, but Older Players Have Inferior Post-Injury Performance

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Word Count: 2911

Tables: 1

Figures: 6

Disclosure Statement: The authors report there are no competing interests to declare.

Running Title: UCL Injuries in NFL Quarterbacks

Author Justification: I affirm that all authors meet the authorship criteria set by the International Committee of Medical Journal Editors (ICMJE). Each author has made substantial contributions to the conception and design of the work, as well as to the acquisition, analysis, and interpretation of data. Additionally, each author has been actively involved in drafting and revising the manuscript, ensuring that the intellectual content is of high quality and relevance. The authors have collectively reviewed and approved the final version of the manuscript to be published. Furthermore, we agree to be accountable for all aspects of the work and are prepared to address any questions related to the accuracy or integrity of the work. We are committed to investigating and resolving any potential issues that may arise, ensuring the work's overall integrity.

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ABSTRACT

Purpose: The purpose of this study was to characterize the epidemiology, mechanism (contact versus non-contact), management, and return to play times for NFL quarterbacks who experienced UCL injuries.

Methods: Using the Pro Sports Transactions Archive, NFL quarterbacks who sustained UCL injuries between 1991 and 2023 were identified. Age at time of injury, mechanism of injury (contact vs non-contact), management strategy, and return to play time (RTP) were recorded. Additionally, player performance metrics including games played, quarterback rating, completion percentage, touchdowns, and interceptions, were examined for the season before injury through the second season post-injury. Player performance analysis was assessed using principal component analysis (PCA), which is a dimensionality reduction statistical method that compresses the several performance metrics into a single value, the first principal component (PC1).

Results: A total of 21 injuries in 20 players were identified. The mean age of quarterbacks at time of injury was 28.7 years (SD = ± 4.6 years). Ten injuries occurred during contact with another player and 11 injuries were non-contact. The majority (13) of UCL injuries were managed non-operatively. Sixteen players achieved RTP, four of which underwent repair or reconstruction (UCLR). Overall, the mean RTP time was 165.6 days (SD = ± 178.8 days), but players who underwent UCLR had a longer average RTP time (359.0 days v. 98.25 days, p = 0.014). Injuries

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sustained after 2006 were associated with improved post injury performance (p = 0.041), but older age at time of injury was associated with diminished post injury performance (p =0.048).

Conclusions: NFL quarterbacks sustain more non-contact UCL injuries and are undergoing UCLR at higher rates than previously reported. Although RTP rates are high and players demonstrate improved post injury performance for injuries sustained after 2006, older age at the time of injury was associated with worse post injury performance.

Level of Evidence: Level IV, Therapeutic Case Series

Key terms: elbow, football (American), NFL, quarterback, ulnar collateral ligament, return to play

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INTRODUCTION

Injury to the ulnar collateral ligament (UCL) of the elbow is a well-documented cause of medial elbow pain and disability in overhead athletes with a predilection towards baseball pitchers.[1–4] The typical mechanism of UCL injury in overhead throwing athletes is repetitive strain causing microtrauma to ligamentous fibers. Patients present with diminished throwing performance (accuracy, velocity, stamina, strength) secondary to posteromedial elbow pain and instability.[5,6] Alternatively, traumatic rupture is often associated with acute pain and inability to throw, with or without an audible "pop."[6] However, American football poses a unique risk profile for UCL injury given the combination of repetitive overhead throwing in addition to traumatic contact secondary to tackling thus increasing the risk of acute injury, albeit via a different mechanism. Dodson et al. reported that 70% of UCL injuries in a subset of professional football quarterbacks were traumatic in nature.[7] Additionally, analysis of the National Collegiate Athletic Association (NCAA) across 25 varsity sports identified three times more elbow UCL injuries in contact athletes compared to strictly throwing athletes.[8] Taken together, the intersection of repetitive overhead throwing and traumatic contact in American football quarterbacks presents an important and underexplored topic that warrants further investigation to enhance our understanding of UCL injuries and their management.

Although UCL injuries can be treated conservatively in most low-demand patients and recreational athletes, UCL reconstruction – colloquially known as 'Tommy John' surgery – has been shown to successfully restore functionality, allowing high-level athletes to return to play (RTP).[9,10] Interestingly, the rate of UCL injury as well as subsequent operative treatment is rising, with epidemiological literature estimating the rate of reconstruction in New York state alone grew by 343% between 2003 and 2014, projecting continued rapid growth through 2025.[1,11,12]

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Currently, there is a paucity of literature discussing UCL injury in National Football League (NFL) quarterbacks. Although existing studies capture a historical subset of this population, the modernization of biologic augmentation techniques such as platelet rich plasma (PRP) injections and UCLR technique warrant a more contemporary review of injury pattern and management strategies.[7,13–18] The purpose of this study was to characterize the epidemiology, mechanism (contact versus non-contact), management, and return to play times for NFL quarterbacks who experienced UCL injuries. The hypotheses for this study are that quarterbacks with UCL injuries would have similar performance metrics following injury, injuries would be mostly traumatic in nature, and would be managed non-operatively with shorter RTP times compared to those managed operatively.

METHODS

Data collection

UCL injuries in NFL quarterbacks were identified from the Pro Sports Transactions Archive which has been used in previous epidemiologic studies in professional sports.[19–23]. Extracted injuries were those that occurred between January 1, 1991 and February 1, 2023, a span of 29 full seasons and approximately one month of a 30th season. Using RStudio 2023.03.2 (RStudio, Boston, MA), player injury data was extracted, including player name, player team, date of injury, notes on injury reserve status, and injured body part. Player injury data was then further filtered for those specifically marked as elbow injuries and in quarterbacks only. Of these results, the authors manually checked each case to ensure it involved an injury to the UCL of the throwing arm. Players sustaining non-throwing UCL injuries were excluded from the analysis. Players identified were then cross referenced and verified using team press releases in conjunction, media

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reports from sources affiliated with the Associated Press, and game logs from NFL-reference.com, which provides data from the official statistics partner of the NBA, Sportradar US (Sportradar AG). Confirmed injuries were then cross referenced with reported injuries from a prior publication on this topic by Dodson et al. in 2010.[7] Information including contact versus non-contact injury, post-injury time to RTP (in days), and whether the player underwent surgery were collected. Time to RTP was defined as return to active roster or full participation in team activities during pre-season practices, and were again verified using team press releases, NFL.com, Associated Press affiliated media reports, and NFL-reference.com, when appropriate. Additionally, pre- and post-injury performance data from Pro Football Reference was collected for each quarterback along with player physical traits, including height, weight, Body Mass Index (BMI), and age at injury. Each player was assigned a numerical value (e.g., Player 1) for de-identification.

Statistical analysis

Power analysis was conducted based on time to RTP. Effect size was determined based on historical data from Dodson et al. as well as relied on the assumption of equal standard deviation between players treated operatively and non-operatively. Because of the limited data available regarding player performance and UCL injury, power analysis could not be calculated for other metrics, and power analysis for time to RTP was applied to these other metrics. Power analysis revealed that at least 18 injuries were needed to achieve a power greater than 0.9 at an effect size of 0.82 for changes in RTP, where we should be able to detect a mean difference of 0.82 of one standard deviation between players of operative versus non-operative treatments. Descriptive statistics including average time from injury to RTP as well as percent of contact versus non-contact injuries were calculated. Player performance analysis was assessed using principal

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component analysis (PCA), which is a dimensionality reduction statistical method that compresses the several performance metrics into a single value, the first principal component (PC1).[24] Multiple orthopedic studies have used PCA as a method to compare groups across multiple metrics including studies investigating sports performance.[24–31] Analysis included the following statistics obtained from Pro Football Reference, for the season prior to injury and through two seasons after injury: games played, Quarterback Rating (QBR), completion percentage, number of touchdowns, and number of interceptions.[32] QBR is a widely used calculation that incorporates five variables including pass attempts, completions, passing yards, touchdown passes, and interceptions.[32] The change in each of the above performance metric was measured as the statistic one season post-injury normalized to performance one season pre-injury. Overall performance via PC1 was first compared for players who underwent UCLR versus those who underwent non-operative management. The PC1 was then compared between players injured before 2005 versus 2006 onwards (independent t-test, alpha = 0.05). Additionally, the PC1 was compared between players injured before 2005 versus 2006 onwards based on operative versus non-operative management (independent t-test, alpha = 0.05). Next, the relationship between age at the time of injury and change in player overall performance was analyzed. Overall performance was again measured using the PC1 from the PCA. The correlation between age at time of injury and PC1 was calculated (Pearson's correlation coefficient, alpha = 0.05). Finally, the distribution of contact versus non-contact injury among older versus younger players was analyzed using Fisher's exact test. The mean age of injury from the sample population was used as a threshold to categorize players as older or younger. Statistical significance for all comparative analyses was set *a priori* at alpha = 0.05.

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All statistics and data visualizations were conducted using Python 3.7 (Python Software Foundation, Wilmington, DE).

RESULTS

We identified 21 UCL injuries in 20 NFL quarterbacks from 1991 to 2023, and the characteristics of each player and injury are described in Table 1. One player suffered a UCL injury twice, and thus the separate injuries were categorized as Player 17a and Player 17b. The average age of quarterbacks at the time of injury was 28.5 years (SD = ± 1.73 years). Ten injuries occurred during contact with another player and 11 injuries were non-contact. Eight players underwent UCLR. Sixteen players returned to play at least one more season in the NFL post-injury. Of the 16 players, nine continued to play at least two more seasons in the NFL post-injury and two players are currently playing in their first season post-injury. Overall, the mean time from injury to RTP was 163.1 days (SD = ± 178.6 days) (Figure 1). There was a significant difference in RTP for players based on management strategy with UCLR averaging 359 days and non-operative management averaging 98.25 days ($p = 0.014$) (Figure 2).

Of the players who returned to play at least one more season, four underwent UCLR and 12 underwent non-operative management. There was no significant difference in change in performance between players who underwent UCLR and those who did not (independent t-test, $p = 0.888$ (Figure 3). The performance of players injured before 2005 versus 2006 and onwards were compared using PCA in Figure 4. Players who were injured 2006 onwards show a statistically significant improvement in performance compared to players injured before 2005 (independent t-test, $p = 0.041$). The subset analysis was conducted based on management type in Figure 5. For the players who underwent UCLR, two players were injured 2005 and before, and two players

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were injured 2006 and onward. Of the players who underwent UCLR, those injured 2006 and onward demonstrated a non-significant improvement in performance compared to those injured 2005 and before (independent t-test, $p = 0.16$). For the players who underwent non-operative management, three players were injured 2005 and before, and two players were injured 2006 and onward. This analysis was similar in that players injured 2006 and onward showed a non-significant improvement in performance (independent t-test, $p = 0.2$).

The distribution of quarterbacks who sustained contact versus non-contact injuries and age were analyzed. Players were categorized according to the mean age (28.7 years old) of the study. Ten players were 29 years or older, and 11 players were younger than 29. Of the players who were above the mean age, five sustained contact injuries, and five sustained non-contact injuries. Of the players who were below the mean age, five sustained contact injuries, and six sustained non-contact injuries. Fisher's exact test revealed no significant difference in the number of contact versus non-contact injuries in the older and younger groups ($p = 1.0$). The relationship between age at the time of injury and change in overall performance one season post-injury versus one season pre-injury was calculated in Figure 6. Overall performance was represented by the PC1, which demonstrated a significant negative correlation with age ($p = 0.048$).

DISCUSSION

The results of the current study identified 21 UCL injuries in 20 NFL quarterbacks from 1991 to 2023 and demonstrated overall high RTP rates, improved post injury performance with injuries sustained after 2006, and inferior post injury performance with older age at the time of injury. While 16 players returned to play with a mean time of 163.1 days, only 9 were able to play for at least 2 seasons post-injury. Injuries were roughly divided equally between both contact (10)

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and non-contact mechanisms (11), and 62% of injuries were managed non operatively. There was no difference in performance based upon management strategy (p= 0.888), but players injured from 2006 and beyond demonstrated superior post-injury performance (p = 0.041). Although not statistically significant, sub-group analysis demonstrated a trend towards improved post-injury performance for both nonoperative (p = 0.2) and UCLR (p = 0.16) for injuries sustained after 2006. Finally, increasing age at the time of injury was associated with worse post injury performance (p = 0.048)

While injuries to the UCL are more commonly described in baseball pitchers, the four-phase American football throw is significantly different from the baseball pitch, with characteristics such as a heavier ball and slower rotational velocity potentially reducing chronic injury rates.[33,34] Although NFL quarterbacks face fewer repetitive stresses than baseball pitchers, they are more prone to contact-related injuries. A study by Dodson et al. covering UCL injuries in NFL quarterbacks from 1994-2008 found that 70% of these injuries were contact-related and nearly all (9/10) were managed nonoperatively. On the contrary, the current study's findings show an equal distribution of contact and non-contact UCL injuries in NFL quarterbacks, suggesting that there may be a more notable impact of repetitive throwing motions than previously believed.[35] Additionally, the lack of correlation between player age and injury mechanism challenges the belief that overuse injuries primarily affect older quarterbacks. Although age did not correlate with the mechanism of injury, there was a negative correlation between age at time of injury and post-injury performance (p = 0.48). This association between age and post-injury performance may prompt a more conservative approach to RTP protocols for older NFL quarterbacks to safeguard their long-term performance. These results reshape our

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understanding of the causes of UCL injury among NFL quarterbacks, and underscores the implications of age at injury on performance and management strategies.

When considering RTP after UCL injury, amongst the 16 players who returned to play, 12 players returned for at least one more season, while 9 played 2 or more post-injury seasons. A significant post-injury performance improvement was noted in post 2006 injuries compared to those sustained in years prior, regardless of management strategy ($p = 0.04$). This period represents a time of expanding interest in biologic therapies in sports medicine, in addition to crucial modifications to the UCL reconstruction techniques, as well as improved rehabilitation protocols.[5,9,10,36,37–45] As such, while it is possible that our results are confounded by heterogeneity of injury severity, they may also reflect the efficacy of post-2006 advances in the treatment of UCL injuries. For example, while elite throwing athletes with incomplete UCL injuries have demonstrated RTP rates of 54-85% at 6-21 weeks following targeted rehabilitation programs,[2,16] multiple relatively recent investigations exploring the role of platelet-rich plasma (PRP) therapy as an adjunct to this protocol have also demonstrated high RTP rates in throwing athletes.[16–18,46,47] A 2024 systematic review and meta-analysis of 8 studies and 322 baseball players by Fucaloro et al demonstrated high return to sport rates when players had rehabilitation programs lasting at least 12 weeks (RTP = 87-100%) and leukocyte-poor PRP preparations (73-100%).[48] While all players in the current study that elected to pursue non-operative management were able RTP, the post-2006 improvements in post-injury performance may represent advances in our understanding of the surgical and nonsurgical management of UCL pathology over the past several decades.

The results of the current study show that greater than 36% of injuries underwent UCLR with a mean RTP of 359 days. Currently, the modified Jobe and Docking techniques are the most

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commonly used UCL reconstruction techniques, and have similar RTP rates ranging from 79.9-97% at approximately 12-18 months.[2,3,14,15] Dodson's original cohort of NFL quarterbacks with complete tears (3/10) were all managed non-operatively with a mean RTP of 67.3 days.[7] While these results suggest an expeditious RTP, interpretation of these findings is limited by the absence of any post-injury performance evaluation. Although not statistically significant, the results of the sub-group analysis demonstrate a trend towards improved post-injury performance for both the nonoperative ($p = 0.2$) and UCLR ($p = 0.16$) groups. Even though these results are limited by the low number of players in each group, this trend may again be reflective of the technical improvements in UCLR procedures that have occurred over the past two decades. Given the results of the current study in the context of the established literature, NFL quarterbacks appear to be undergoing UCLR at a rate higher than previously established.

While RTP rates are important, no analysis on elite athletic injuries is complete without an assessment of athletic performance following RTP. The current study provides a nuanced understanding of the injury's implications on quarterback performance, highlighting that while historically many players successfully RTP, their post-injury performance levels are impacted by age at the time of injury and year of injury. These findings suggest that RTP, while an important milestone, does not fully encapsulate the injury's long-term impact on an athlete's career. Our findings indicate that the quality of performance post-injury, particularly in a high-impact sport like professional football, may be more difficult to predict than previously thought. Such insights are crucial not only for clinicians and athletic trainers but also for players and team management in strategizing long-term player development and team composition in the wake of such injuries.

Limitations

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Several limitations must be acknowledged. The retrospective nature of the study and reliance on publicly available data sources inherently limits the depth of injury specifics and treatment details we could access. This reliance on secondary data sources may lead to underreporting of injuries or incomplete data regarding the full spectrum of treatment modalities employed, particularly for grading of injury severity, duration of rehabilitation, use of biological augments such as PRP, and surgical techniques. Secondly, our analysis was confined to NFL quarterbacks, a highly specialized and elite athlete population. Therefore, the findings may not be generalizable to other athlete populations, including quarterbacks at different levels of play or athletes in other sports where throwing mechanics and injury risks differ. As a byproduct of this highly specialized group, the small sample size limits the statistical power for some of the analyses and may contribute to variability in the results as this increases the probability of encountering type II error. Furthermore, our analysis of player performance post-injury, though rigorous, did not account for all possible confounding factors, such as changes in team dynamics, coaching strategies, or personal factors affecting the athletes' performance.

CONCLUSION

NFL quarterbacks sustain more non-contact UCL injuries and are undergoing UCLR at higher rates than previously reported. Although RTP rates are high and players demonstrate improved post injury performance for injuries sustained after 2006, older age at the time of injury was associated with worse post injury performance.

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REFERENCES

1. Petty DH, Andrews JR, Fleisig GS, Cain EL. Ulnar collateral ligament reconstruction in high school baseball players: clinical results and injury risk factors. *Am J Sports Med*. 2004;32(5):1158-1164. doi:10.1177/0363546503262166
2. Ford GM, Genuario J, Kinkartz J, Githens T, Noonan T. Return-to-Play Outcomes in Professional Baseball Players After Medial Ulnar Collateral Ligament Injuries: Comparison of Operative Versus Nonoperative Treatment Based on Magnetic Resonance Imaging Findings. *Am J Sports Med*. 2016;44(3):723-728. doi:10.1177/0363546515621756
3. Thomas SJ, Paul RW, Rosen AB, et al. Return-to-Play and Competitive Outcomes After Ulnar Collateral Ligament Reconstruction Among Baseball Players: A Systematic Review. *Orthop J Sports Med*. 2020;8(12):2325967120966310. doi:10.1177/2325967120966310
4. Walker CM, Genuario JW, Houck DA, Murayama S, Mendez H, Noonan TJ. Return-to-Play Outcomes in Professional Baseball Players After Nonoperative Treatment of Incomplete Medial Ulnar Collateral Ligament Injuries: A Long-Term Follow-up Study. *Am J Sports Med*. 2021;49(5):1137-1144. doi:10.1177/0363546521996706
5. Chen FS, Rokito AS, Jobe FW. Medial Elbow Problems in the Overhead-Throwing Athlete. J Am Acad Orthop Surg . 2001;9(2):99.
6. Cain EL, Dugas JR, Wolf RS, Andrews JR. Elbow Injuries in Throwing Athletes: A Current Concepts Review. *Am J Sports Med*. 2003;31(4):621-635. doi:10.1177/03635465030310042601
7. Dodson CC, Slenker N, Cohen SB, Ciccotti MG, DeLuca P. Ulnar collateral ligament injuries of the elbow in professional football quarterbacks. J Shoulder Elbow Surg.2010;19(8):1276-1280. doi:10.1016/j.jse.2010.05.028
8. Li NY, Goodman AD, Lemme NJ, Owens BD. Epidemiology of Elbow Ulnar Collateral Ligament Injuries in Throwing Versus Contact Athletes of the National Collegiate Athletic Association: Analysis of the 2009-2010 to 2013-2014 Seasons. *Orthop J Sports Med*. 2019;7(4):2325967119836428. doi:10.1177/2325967119836428
9. Vitale MA, Ahmad CS. The outcome of elbow ulnar collateral ligament reconstruction in overhead athletes: a systematic review. *Am J Sports Med*. 2008;36(6):1193-1205. doi:10.1177/0363546508319053

Journal Pre-proof

10. Langer P, Fadale P, Hulstyn M. Evolution of the treatment options of ulnar collateral ligament injuries of the elbow. *Br J Sports Med*. 2006;40(6):499-506. doi:10.1136/bjsm.2005.025072
11. Hodgins JL, Vitale M, Arons RR, Ahmad CS. Epidemiology of Medial Ulnar Collateral Ligament Reconstruction: A 10-Year Study in New York State. *Am J Sports Med*. 2016;44(3):729-734. doi:10.1177/0363546515622407
12. Mahure SA, Mollon B, Shamah SD, Kwon YW, Rokito AS. Disproportionate trends in ulnar collateral ligament reconstruction: projections through 2025 and a literature review. *J Shoulder Elbow Surg*. 2016;25(6):1005-1012. doi:10.1016/j.jse.2016.02.036
13. Kenter K, Behr CT, Warren RF, O'Brien SJ, Barnes R. Acute elbow injuries in the National Football League. *J Shoulder Elbow Surg*. 2000;9(1):1-5. doi:10.1016/s1058-2746(00)80023-3
14. Arner JW, Chang ES, Bayer S, Bradley JP. Direct Comparison of Modified Jobe and Docking Ulnar Collateral Ligament Reconstruction at Midterm Follow-up. *Am J Sports Med*. 2019;47(1):144-150. doi:10.1177/0363546518812421
15. Yoshida M, Takenaga T, Takeuchi S, et al. Ulnar Collateral Ligament Reconstruction With the Modified Jobe Technique in Baseball Pitchers. *Orthop J Sports Med*. 2021;9(4):2325967121998339. doi:10.1177/2325967121998339
16. Chauhan A, McQueen P, Chalmers PN, et al. Nonoperative Treatment of Elbow Ulnar Collateral Ligament Injuries With and Without Platelet-Rich Plasma in Professional Baseball Players: A Comparative and Matched Cohort Analysis. *Am J Sports Med*. 2019;47(13):3107-3119. doi:10.1177/0363546519876305
17. Deal JB, Smith E, Heard W, O'Brien MJ, Savoie FH. Platelet-Rich Plasma for Primary Treatment of Partial Ulnar Collateral Ligament Tears: MRI Correlation With Results. *Orthopaedic Journal of Sports Medicine*. 2017;5(11):232596711773823. doi:10.1177/2325967117738238
18. Dines JS, Williams PN, ElAttrache N, et al. Platelet-Rich Plasma Can Be Used to Successfully Treat Elbow Ulnar Collateral Ligament Insufficiency in High-Level Throwers. *Am J Orthop (Belle Mead NJ)*. 2016;45(5):296-300.
19. Pro Sports Transactions Archive. Accessed June 27, 2023. https://www.prosportstransactions.com/

Journal Pre-proof

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20. Morikawa LH, Tummala SV, Brinkman JC, Buckner Petty SA, Chhabra A. Effect of a Condensed NBA Season on Injury Risk: An Analysis of the 2020 Season and Player Safety. *Orthop J Sports Med*. 2022;10(9):23259671221121116. doi:10.1177/23259671221121116
21. Lin E, Tummala SV, Morikawa L, et al. Strains/Sprains and Fractures Are the Most Common Hand and Wrist Injuries in National Basketball Association Athletes Who Return to Preinjury Player Efficiency and Equal or Greater True Shooting Percentage Within Two Years of Injury. *Arthrosc Sports Med Rehabil*. 2023;5(6):100829. doi:10.1016/j.asmr.2023.100829
22. Bullock GS, Ferguson T, Arundale AH, Martin CL, Collins GS, Kluzek S. Return to performance following severe ankle, knee, and hip injuries in National Basketball Association players. *PNAS Nexus*. 2022;1(4):pgac176. doi:10.1093/pnasnexus/pgac176
23. Tummala SV, Morikawa L, Brinkman JC, et al. Characterization of Ankle Injuries and Associated Risk Factors in the National Basketball Association: Minutes Per Game and Usage Rate Associated With Time Loss. *Orthop J Sports Med*. 2023;11(7):23259671231184459. doi:10.1177/23259671231184459
24. Zhang Z, Castelló A. Principal components analysis in clinical studies. *Ann Transl Med*. 2017;5(17):351. doi:10.21037/atm.2017.07.12
25. McCormack S, Jones B, Scantlebury S, Collins N, Owen C, Till K. Using Principal Component Analysis to Compare the Physical Qualities Between Academy and International Youth Rugby League Players. *Int J Sports Physiol Perform*. 2021;16(12):1880-1887. doi:10.1123/ijspp.2021-0049
26. Goreham JA, Landry SC, Kozey JW, Smith B, Ladouceur M. Using principal component analysis to investigate pacing strategies in elite international canoe kayak sprint races. *Sports Biomech*. 2023;22(11):1444-1459. doi:10.1080/14763141.2020.1806348
27. Foster BH, Shaw CB, Boutin RD, et al. A principal component analysis-based framework for statistical modeling of bone displacement during wrist maneuvers. *J Biomech*. 2019;85:173-181. doi:10.1016/j.jbiomech.2019.01.030
28. Daruwalla ZJ, Courtis P, Fitzpatrick C, Fitzpatrick D, Mullett H. An application of principal component analysis to the clavicle and clavicle fixation devices. *J Orthop Surg Res*. 2010;5:21. doi:10.1186/1749-799X-5-21
29. Shalhoub S, Cyr A, Maletsky LP. Correlation between knee anatomy and joint laxity using

principal component analysis. *J Orthop Res*. 2022;40(11):2502-2509. doi:10.1002/jor.25294

30. Welch N, Richter C, Moran K, Franklyn-Miller A. Principal Component Analysis of the Associations Between Kinetic Variables in Cutting and Jumping, and Cutting Performance Outcome. *J Strength Cond Res*. 2021;35(7):1848-1855. doi:10.1519/JSC.0000000000003028
31. Mantovani G, Lamontagne M, Varin D, Cerulli GG, Beaulé PE. Comparison of total hip arthroplasty surgical approaches by Principal Component Analysis. *J Biomech*. 2012;45(12):2109-2115. doi:10.1016/j.jbiomech.2012.05.041
32. Pro Football Stats, History, Scores, Standings, Playoffs, Schedule & Records. Pro-Football-Reference.com. Accessed June 27, 2023. https://www.pro-football-reference.com/
33. Fleisig GS, Escamilla RF, Andrews JR, Matsuo T, Satterwhite Y, Barrentine SW. Kinematic and Kinetic Comparison between Baseball Pitching and Football Passing. *Journal of Applied Biomechanics*. 1996;12(2):207-224. doi:10.1123/jab.12.2.207
34. Kelly BT, Backus SI, Warren RF, Williams RJ. Electromyographic analysis and phase definition of the overhead football throw. *Am J Sports Med*. 2002;30(6):837-844. doi:10.1177/03635465020300061401
35. Erickson BJ, Hurley ET, Mojica ES, Jazrawi LM, Ulnar Collateral Ligament – Delphi Consensus Group (UCL-DCG). Elbow Ulnar Collateral Ligament Tears: A Modified Consensus Statement. *Arthroscopy*. 2023;39(5):1161-1171. doi:10.1016/j.arthro.2022.12.033
36. Paletta GA, Wright RW. The Modified Docking Procedure for Elbow Ulnar Collateral Ligament Reconstruction: 2-Year Follow-up in Elite Throwers. *Am J Sports Med*. 2006;34(10):1594-1598. doi:10.1177/0363546506289884
37. Murray MM, Spindler KP, Abreu E, et al. Collagen-platelet rich plasma hydrogel enhances primary repair of the porcine anterior cruciate ligament. *J Orthop Res*. 2007;25(1):81-91. doi:10.1002/jor.20282
38. Han J, Meng HX, Tang JM, Li SL, Tang Y, Chen ZB. The effect of different platelet-rich plasma concentrations on proliferation and differentiation of human periodontal ligament cells *in vitro*. *Cell Proliferation*. 2007;40(2):241-252. doi:10.1111/j.1365-2184.2007.00430.x
39. Kawase T, Okuda K, Saito Y, Yoshie H. In Vitro Evidence That the Biological Effects of Platelet-Rich Plasma on Periodontal Ligament Cells Is Not Mediated Solely by Constituent Transforming-Growth Factor-β or Platelet-Derived Growth Factor. *Journal of Periodontology*. 2005;76(5):760-767. doi:10.1902/jop.2005.76.5.760

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40. Smith JJ, Ross MW, Smith RKW. Anabolic effects of acellular bone marrow, platelet rich plasma, and serum on equine suspensory ligament fibroblasts in vitro. *Vet Comp Orthop Traumatol*. 2006;19(1):43-47.
41. Dines JS, ElAttrache NS, Conway JE, Smith W, Ahmad CS. Clinical Outcomes of the DANE TJ Technique to Treat Ulnar Collateral Ligament Insufficiency of the Elbow. *Am J Sports Med*. 2007;35(12):2039-2044. doi:10.1177/0363546507305802
42. Dodson CC, Thomas A, Dines JS, Nho SJ, Williams RJ, Altchek DW. Medial Ulnar Collateral Ligament Reconstruction of the Elbow in Throwing Athletes. *Am J Sports Med*. 2006;34(12):1926-1932. doi:10.1177/0363546506290988
43. Hall MP, Band PA, Meislin RJ, Jazrawi LM, Cardone DA. Platelet-rich plasma: current concepts and application in sports medicine. *J Am Acad Orthop Surg*. 2009;17(10):602-608. doi:10.5435/00124635-200910000-00002
44. Virchenko O, Aspenberg P. How can one platelet injection after tendon injury lead to a stronger tendon after 4 weeks? Interplay between early regeneration and mechanical stimulation. *Acta Orthop*. 2006;77(5):806-812. doi:10.1080/17453670610013033
45. Rohrbough JT, Altchek DW, Hyman J, Williams RJ, Botts JD. Medial Collateral Ligament Reconstruction of the Elbow using the Docking Technique. *Am J Sports Med*. 2002;30(4):541-548. doi:10.1177/03635465020300041401
46. Podesta L, Crow SA, Volkmer D, Bert T, Yocum LA. Treatment of Partial Ulnar Collateral Ligament Tears in the Elbow With Platelet-Rich Plasma. *Am J Sports Med*. 2013;41(7):1689-1694. doi:10.1177/0363546513487979
47. Conant BJ, German NA, David SL. The Use of Platelet-Rich Plasma for Conservative Treatment of Partial Ulnar Collateral Ligament Tears in Overhead Athletes: A Critically Appraised Topic. *J Sport Rehabil*. 2020;29(4):509-514. doi:10.1123/jsr.2018-0174
48. Fucaloro SP, Kang AS, Bragg JT, Krivicich L, Salzler MJ. Return to Sport After Ulnar Collateral Ligament Tears Treated with Platelet-Rich Plasma Injections is Influenced by Length of Rehabilitation and Leukocyte Content of Injections: A Systematic Review. *Arthroscopy*. Published online March 18, 2024:S0749-8063(24)00205-6. doi:10.1016/j.arthro.2024.03.017

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TABLES

Table 1. Player demographics, injury characteristics, treatment plan, and time for RTP in days. *N/A, player did not play in an NFL game after injury; UCLR, ulnar collateral ligament reconstruction or repair; RTP, return to play; Non-op, non-operative management.

Player	Age at time of injury	Year	Mechanism	Treatment	Days to RTP
1	35	1991	Non-contact	UCLR	446
2	25	1994	Contact	Non-op	5
3	29	1998	Non-contact	UCLR	N/A
4	28	2001	Non-contact	Non-op	30
5	36	2001	Non-contact	UCLR	410
6	30	2003	Non-contact	UCLR	N/A
7	33	2004	Contact	Non-op	42
8	29	2004	Contact	Non-op	14
9	24	2005	Non-contact	Non-op	22
10	32	2007	Non-contact	UCLR	350
11	28	2008	Contact	Non-op	357
12	33	2008	Contact	Non-op	70
13	38	2008	Contact	Non-op	1
14	29	2008	Contact	Non-op	131

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15	26	2013	Non-contact	UCLR	N/A
16	25	2016	Non-contact	UCLR	N/A
17a	22	2018	Non-contact	Non-op	42
18	25	2019	Non-contact	Non-op	458
19	25	2020	Contact	Non-op	365
17b	26	2022	Contact	Non-op	7
20	23	2023	Contact	UCLR	224

Figure Legend

Figure 1. Boxplot visualizing the number of days between a player's injury and return to play (RTP). Players who did not return to play were excluded from the plot. The 25th percentile is 20.0 days, the median is 56.0 days, and the 75th percentile is 357.0 days. The mean RTP is 163.1 days. RTP, return to play.

Figure 2. Comparing mean time from injury to return to play (RTP) among players who underwent UCLR versus those who did not (non-parametric t-test, p=0.014). Mean RTP for players who underwent UCLR was 359.0 days. Mean RTP for players who underwent non-operative management was 98.25 days. UCLR, ulnar collateral ligament reconstruction.

Figure 3. Comparison of mean PC1 of players who underwent UCLR and those who did not. Principal component analysis was used to analyze the change in performance one season before

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injury to one season post injury of the following: number of games played, number of touchdowns, number of interceptions, QBR, completion percentage, and number of passing yards. A greater PC1 indicates an improved performance compared to the season before injury; a lesser PC1 indicates a worse performance. An independent t-test was used to compare the PC1 ($p = 0.888$).

Figure 4. Comparison of mean PC1 of players injured before 2005 versus 2006 onwards. Principal component analysis was used to analyze the change in performance one season before injury to one season post injury of the following: number of games played, number of touchdowns, number of interceptions, QBR, completion percentage, and number of passing yards. The PC1 is a statistical representation of all six performance metrics. A greater PC1 indicates an improved performance compared to the season before injury; a lesser PC1 indicates a worse performance. An independent t-test was used to compare the PC1 ($p = 0.04$).

Figure 5. Analyzing the performance of players, subdivided by management, comparing mean PC1 of those who were injured before 2005 versus 2006 and onward. Principal component analysis was used to analyze the change in performance one season before injury to one season post injury of the following: number of games played, number of touchdowns, number of interceptions, QBR, completion percentage, and number of passing yards. A greater PC1 indicates an improved performance compared to the season before injury; a lesser PC1 indicates a worse performance. An independent t-test was used to compare the PC1 (non- operative: $p = 0.2$, UCLR: $p = 0.16$).

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Figure 6. Assessing the relationship between age at time of injury and the change in player performance before and after injury. Principal component analysis was used to analyze the change in performance one season before injury to one season post injury of the following: number of games played, number of touchdowns, number of interceptions, QBR, completion percentage, and number of passing yards. Change in performance metric was determined by normalizing the metric from the first season after injury to that from the season before injury ($p = 0.048$)

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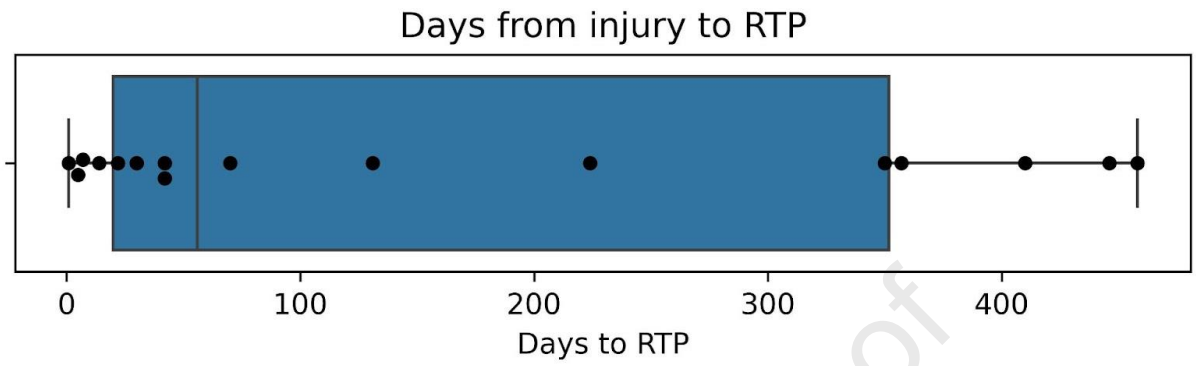
FIGURES



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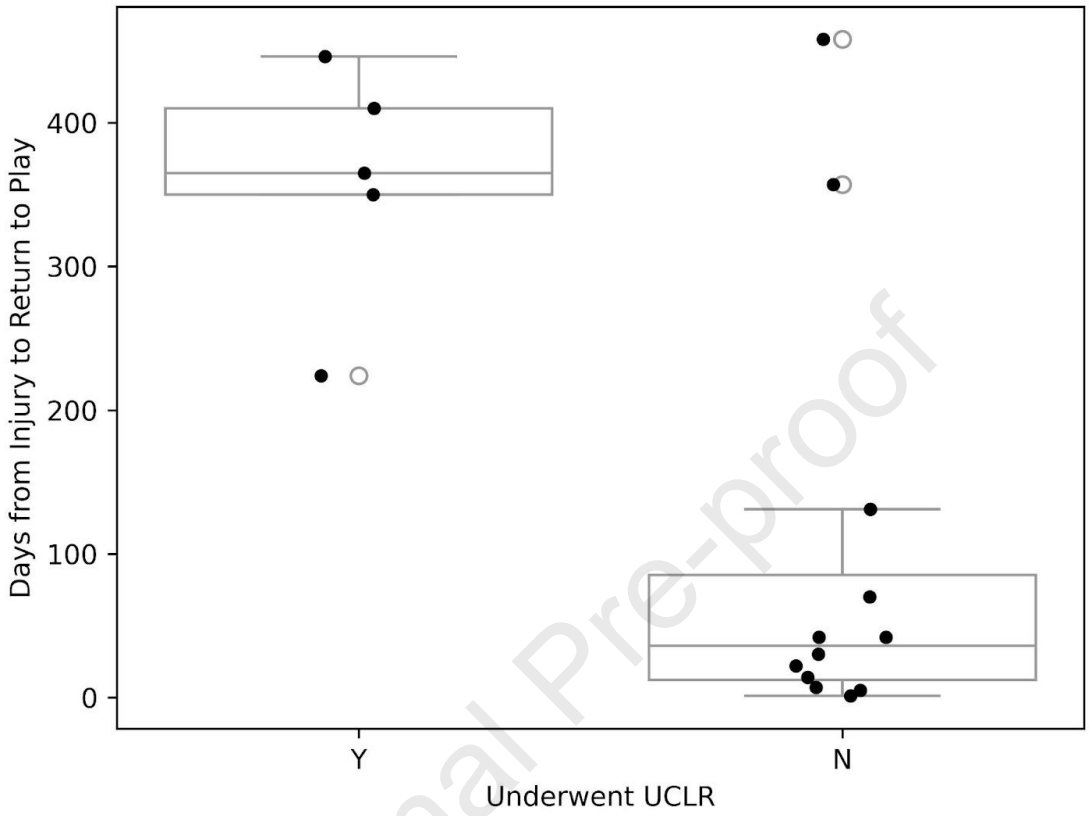

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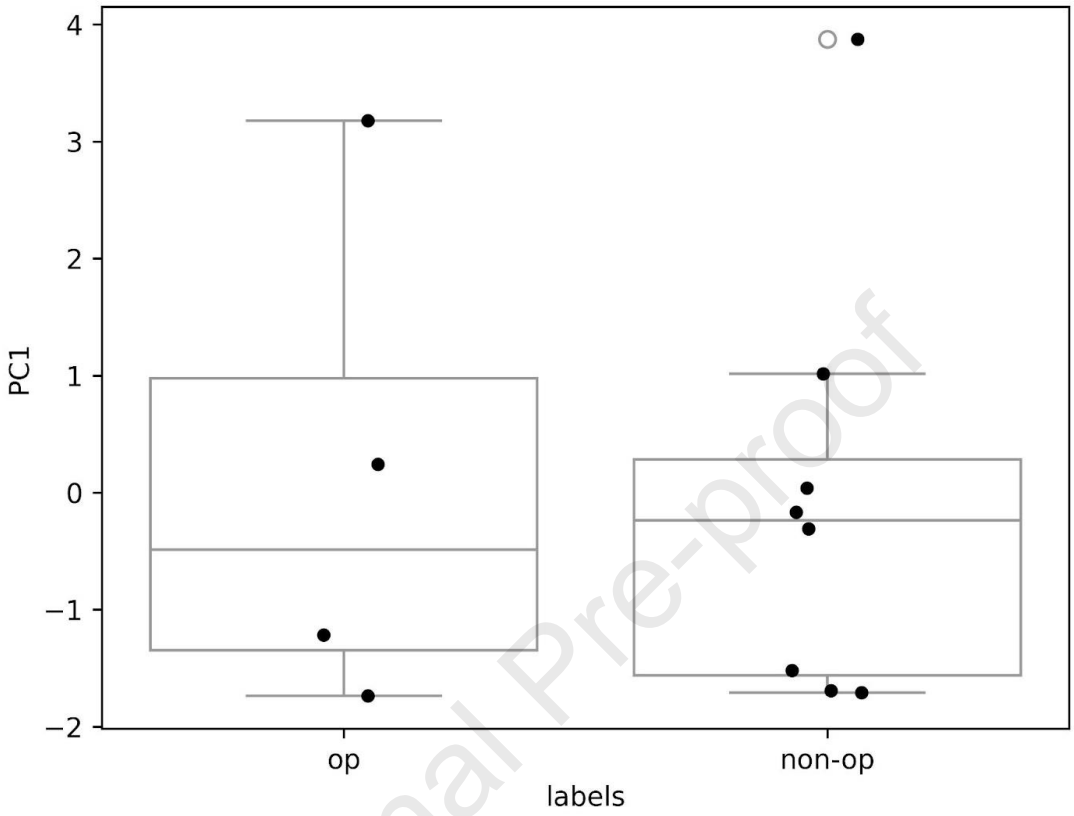

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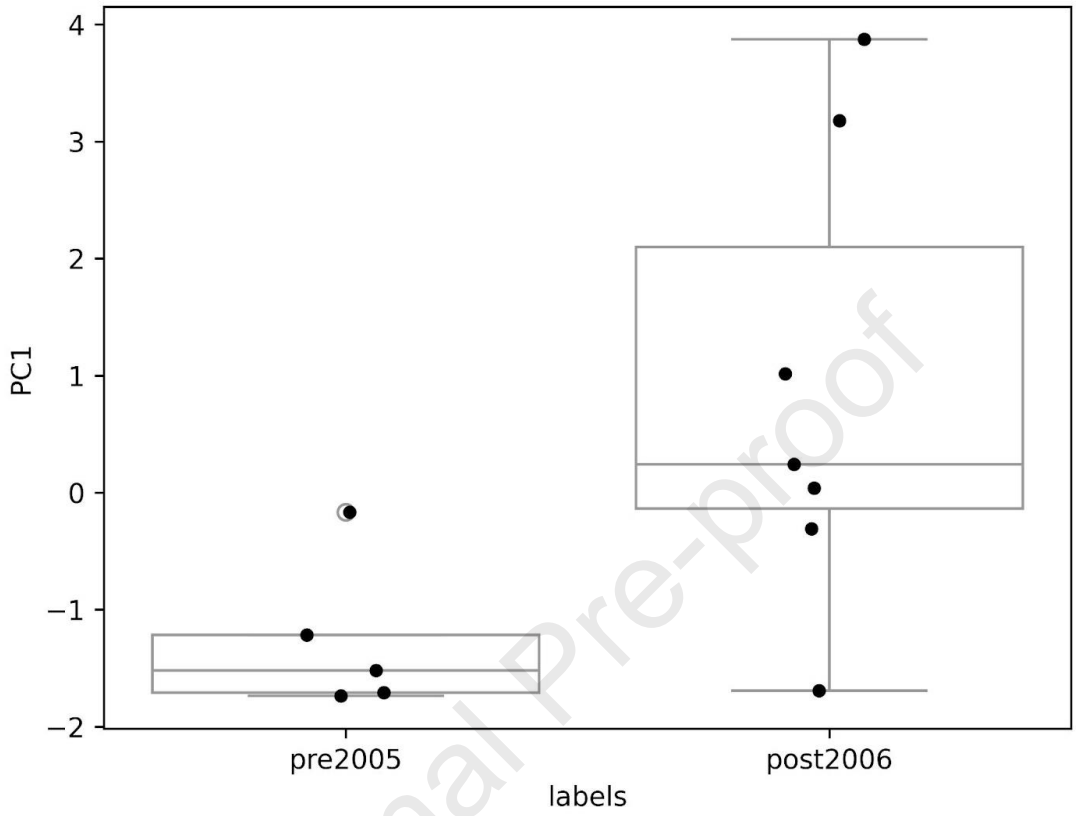

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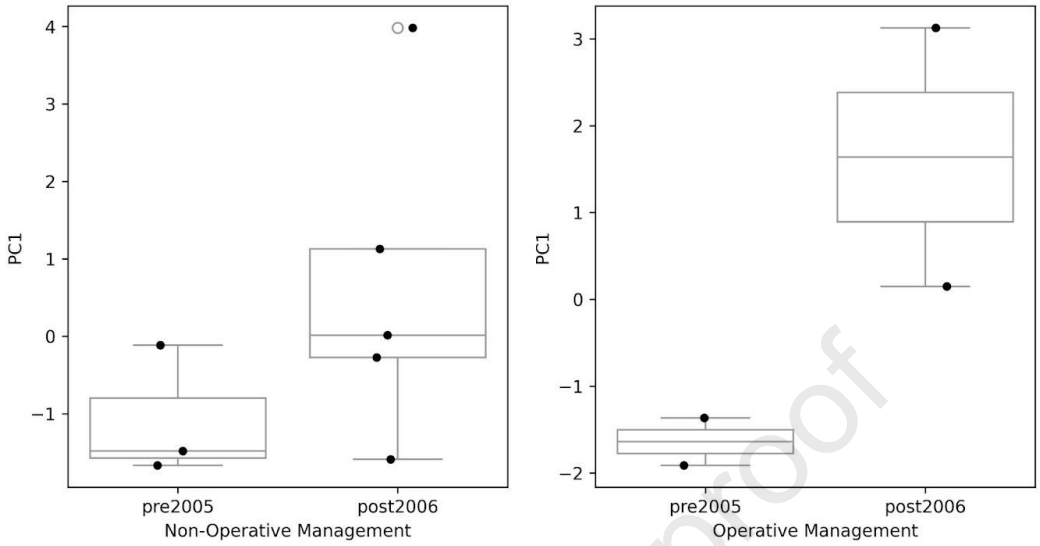

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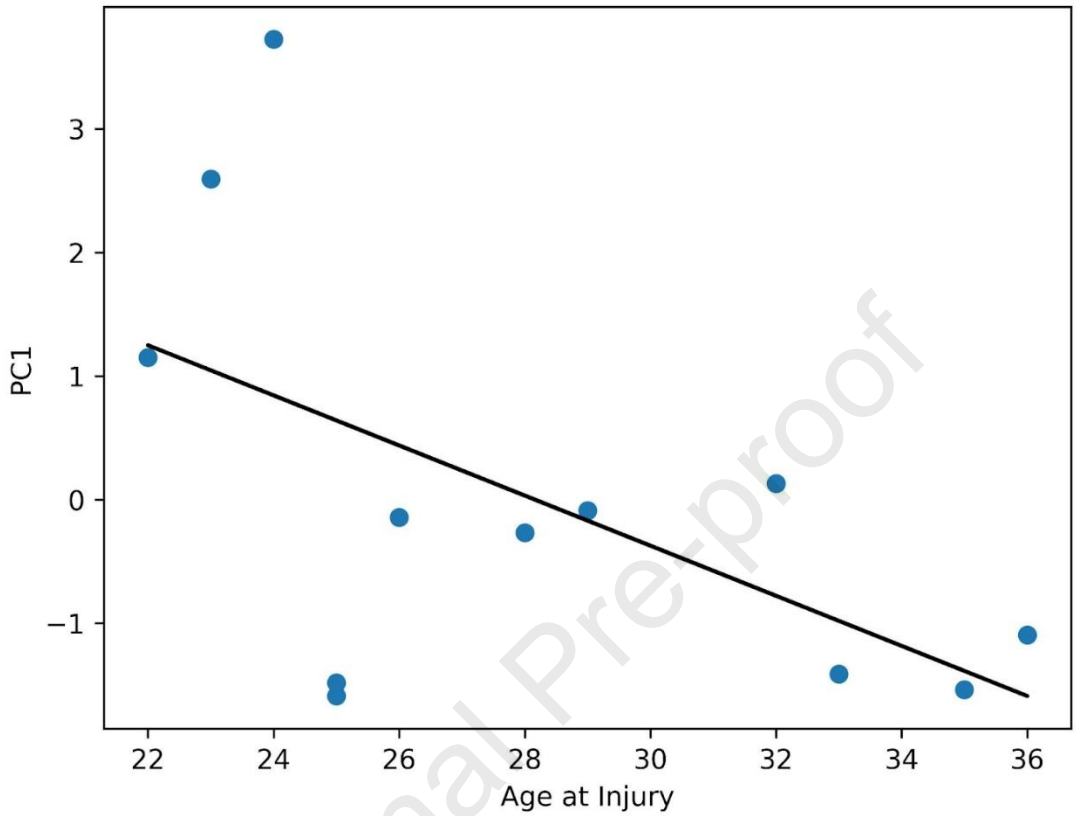

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